

SegmentIQ

Customer Segmentation Analysis Report

Report Generated:	March 23, 2026 at 22:25
Total Customers Analysed:	100
Best Algorithm:	DBSCAN
Segments Identified:	5
Features Used:	11

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Executive Summary

This report presents the results of an AI-powered customer segmentation analysis performed on **100 customers** using multiple clustering algorithms. The analysis identified **5 distinct customer segments**, each with unique behavioural patterns, value profiles, and engagement characteristics. The best performing algorithm was **DBSCAN** based on silhouette score evaluation. Each segment has been profiled with tailored business recommendations to drive targeted marketing, loyalty initiatives, and revenue growth strategies.

Algorithm Performance Comparison

Algorithm	Clusters	Silhouette ↑	Davies-Bouldin ↓	Calinski-Harabasz ↑
K-Means	5	0.6354	0.565	362.72
DBSCAN ✓	5	0.8212	0.2458	2062.67
Hierarchical	5	0.6354	0.565	362.72
GMM	5	0.6423	0.5356	354.06

Segment Overview

Segment	Size	% Share	Priority Action
■ Steady Voyagers	11	11.0%	Grow & Upsell
■ Steady Voyagers	9	9.0%	Grow & Upsell
■ Steady Voyagers	5	5.0%	Grow & Upsell
■ Platinum Loyalists	9	9.0%	Retain & Reward
■ Steady Voyagers	10	10.0%	Grow & Upsell

Detailed Segment Profiles

■ Segment 0: Steady Voyagers

Size: 11 customers (11.0% of total) | **Priority:** Grow & Upsell

Regular customers with moderate spend — the backbone segment.

Age	25.5
Annual Income	33090.9
Total Trips	1.2
Avg Spend Per Trip	823.6
Loyalty Years	1.0
Membership Tier	2.0

Recommended Actions:

- Offer group booking discounts to increase trip frequency
- Cross-sell onboard packages (spa, dining, excursions)
- Introduce referral programme with travel credits
- Seasonal promotions aligned to booking history
- Nudge towards next loyalty tier with progress tracker

■ Segment 1: Steady Voyagers

Size: 9 customers (9.0% of total) | **Priority:** Grow & Upsell

Regular customers with moderate spend — the backbone segment.

Age	38.0
Annual Income	71777.8
Total Trips	8.4
Avg Spend Per Trip	4066.7
Loyalty Years	5.0
Membership Tier	0.0

Recommended Actions:

- Offer group booking discounts to increase trip frequency
- Cross-sell onboard packages (spa, dining, excursions)
- Introduce referral programme with travel credits
- Seasonal promotions aligned to booking history
- Nudge towards next loyalty tier with progress tracker

■ Segment 2: Steady Voyagers

Size: 5 customers (5.0% of total) | **Priority:** Grow & Upsell

Regular customers with moderate spend — the backbone segment.

Age	33.8
Annual Income	58800.0
Total Trips	5.2
Avg Spend Per Trip	2600.0
Loyalty Years	3.0
Membership Tier	2.0

Recommended Actions:

- Offer group booking discounts to increase trip frequency
 - Cross-sell onboard packages (spa, dining, excursions)
 - Introduce referral programme with travel credits
 - Seasonal promotions aligned to booking history
 - Nudge towards next loyalty tier with progress tracker
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■ **Segment 3: Platinum Loyalists**

Size: 9 customers (9.0% of total) | **Priority:** Retain & Reward

High-value, long-term customers with excellent engagement.

Age	56.0
Annual Income	115666.7
Total Trips	21.0
Avg Spend Per Trip	9155.6
Loyalty Years	13.7
Membership Tier	1.0

Recommended Actions:

- Offer exclusive suite upgrades and priority boarding
 - Invite to VIP loyalty events and pre-launch sailings
 - Provide dedicated concierge service
 - Early access to new route announcements
 - Personalised anniversary/milestone rewards
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■ **Segment 4: Steady Voyagers**

Size: 10 customers (10.0% of total) | **Priority:** Grow & Upsell

Regular customers with moderate spend — the backbone segment.

Age	30.2
Annual Income	45100.0
Total Trips	3.1
Avg Spend Per Trip	1575.0

Loyalty Years	1.9
Membership Tier	2.0

Recommended Actions:

- Offer group booking discounts to increase trip frequency
 - Cross-sell onboard packages (spa, dining, excursions)
 - Introduce referral programme with travel credits
 - Seasonal promotions aligned to booking history
 - Nudge towards next loyalty tier with progress tracker
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Feature Importance Analysis

Features ranked by combined importance score across PCA loadings, Mutual Information, and Random Forest importance.

Rank	Feature	Importance Score
1	Avg Spend Per Trip	0.1705
2	Annual Income	0.1671
3	Onboard Spend	0.1656
4	Referrals Made	0.1583
5	Loyalty Years	0.1565
6	Age	0.1533
7	Total Trips	0.1525
8	Days Since Last Trip	0.1287
9	Num Complaints	0.1202
10	Cancelled Trips	0.1190

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